

Isochrone overturning in steady glacier flow

A cumulative reverse-shear criterion for full-Stokes flow

Bradley Paul Lipovsky

Department of Earth and Space Sciences, University of Washington, Seattle, USA

Correspondence: bp17@uw.edu

Abstract

Ancient ice recovered at Allan Hills extends into the Miocene, but steep layers, age reversals and small-scale folds complicate its interpretation. Existing steady-flow theory shows that isochrones cannot overturn where horizontal speed increases upward. Full-Stokes flow over rough beds can contain local extrusion, so this pointwise condition is stronger than necessary. Here, we derive a signed order factor that measures whether particles deposited in sequence retain their horizontal order along an isochrone. Stratigraphic order depends on the cumulative competition between normal and reverse vertical shear, weighted most strongly where horizontal motion is slow. Reverse shear must erase both the ordering imposed at deposition and the accumulated normal shear before a layer becomes vertical. Full-Stokes mechanics provides no universal sign guarantee, although shallow-ice flow preserves order. We derive a full-Stokes threshold, bounds for non-stagnant localized obstacles, a first-order cancellation result for periodic roughness about plug flow and a trajectory diagnostic. These results distinguish topographic draping from overturning at Allan Hills and other old-ice sites.

Keywords: ice stratigraphy; recumbent folds; Allan Hills; NSF COLDEX; full-Stokes flow; extrusion flow; age equation; blue ice

1 Introduction

Recovering old ice is useful only where its age and stratigraphic context can be established. The European Beyond EPICA–Oldest Ice project used radar stratigraphy and ice-flow modelling at Little Dome C to target a continuous record extending beyond 1 Ma (Lilien et al., 2021; Chung et al., 2025). Antarctic blue-ice areas provide a complementary archive: surface mass loss and upward ice motion expose discontinuous climatic intervals that can be much older than conventional vertical cores (Bintanja, 1999; Spaulding et al., 2012).

At Allan Hills, shallow cores first yielded a disturbed section near 1 Ma (Higgins et al., 2015), then atmospheric snapshots older than 2 Ma (Yan et al., 2019), and most recently Pliocene and Miocene ice and air with reported ages approaching 6 Ma (Shackleton et al., 2025). These cores are a focus of NSF COLDEX, which seeks to reconstruct atmospheric composition and Antarctic climate beyond the continuous 800 ka ice-core record.

These records are stratigraphically difficult to interpret. Allan Hills ice is strongly thinned and locally out of order. The ALHIC1901 age–depth relation includes multiple reversals, while the deepest ice can represent very large time intervals per unit core length (Shackleton et al., 2025). Three-dimensional electrical conductivity measurements resolve mean layer dips of approximately 29° and 69° in two recent Allan Hills cores (Kirkpatrick et al., 2025). Microstructural observations from ALHIC1901 indicate pervasive simple shear, strongly elongated bubbles, crystal-orientation stripes, and centimetre-scale kink and z-folds (Stoll et al., 2026). Radar and flow modelling independently show that the duration and continuity of the archive vary sharply across the blue-ice area and with basal geometry (Kehrl et al., 2018). A recent three-dimensional full-Stokes model predicts near-stagnant basal ice upstream of the local nunatak and faster flow into the Cul-de-sac from a distinct accumulation environment (Manos et al., 2026). Deformation and transport history therefore control which ages are present, how they are ordered and how much time a sample averages.

A basic unresolved question is whether a fixed bed obstacle in an otherwise steady glacier can create the finite-amplitude disturbance required for a recumbent fold. Flow over topography certainly bends internal layers. Full mechanical calculations predict draping over long-wavelength topography and overriding of shorter-wavelength relief (Hindmarsh et al., 2006). Rough-bed Stokes solutions can also contain local extrusion, meaning that horizontal speed increases downward over part of the ice column (Gudmundsson, 1997a; Waddington, 2010). At larger bed slopes, separated secondary circulation is possible even in creeping flow (Gudmundsson, 1997b). These results appear, at first sight, to undermine the common intuition that a steady obstacle can produce only a standing layer disturbance rather than a recumbent fold.

These statements concern different parts of the layer geometry. A material tangent is an infinitesimal line segment joining neighboring particles within a layer. Flow around an obstacle can bend the layer and rotate this segment locally. The layer remains ordered, however, if particles deposited at successively larger upstream positions also occur at successively larger downstream positions everywhere along an equal-age surface. A recumbent fold begins only where that horizontal ordering vanishes and then reverses. Curvature and local tangent rotation can therefore occur without an order reversal.

The local theory of Waddington et al. (2001), hereafter WBA, formalized how a pre-existing wrinkle is flattened by longitudinal extension or overturned by shear. WBA also stated the key qualitative principle that layers spending their entire history in a

65 steady-state flow field are not passively folded; a transient process must first create a disturbance relative to the steady stratigraphy. Jacobson and Waddington (2004) replaced the local criterion with finite-strain integration along trajectories, and Jacobson and Waddington (2005) showed that divide migration can place an inherited Raymond arch in a shear regime that turns it into a recumbent fold. For steady stratigraphy, Parrenin et al. 70 (2006) introduced logarithmic flux coordinates for analytical age solutions, and Parrenin and Hindmarsh (2007) generalized the construction to horizontally non-uniform velocity profiles. Their Appendix A showed that an isochrone cannot pass through vertical when horizontal velocity increases monotonically upward. Corrections for flow-induced layer thinning date at least to Nye (1963) and the early depth–age model of Dansgaard and 75 Johnsen (1969). The same characteristic framework subsequently underpinned numerical age solvers (Hindmarsh et al., 2009; Parrenin et al., 2025) and an explicit vertical thinning function (Parrenin, 2013; Parrenin et al., 2025).

Monotonic upward speed is sufficient but not necessary for preserving order because full-Stokes glacier flow does not impose $u_z \geq 0$ pointwise. The relevant quantity is 80 cumulative: how much reverse shear can a particle experience before the depositional ordering of the isochrone is lost? We answer this question for steady two-dimensional flow by:

- (i) derive a physical-coordinate representation of the signed inverse vertical-thinning factor and its interpretation as horizontal order along an isochrone;
- 85 (ii) express its sign as a cumulative competition between surface-fast and reverse vertical shear;
- (iii) decompose the reverse-shear budget using the full-Stokes stress balance;
- (iv) identify regimes in which ordinary glacier mechanics guarantees order and regimes in which it does not;
- 90 (v) derive a sufficient no-overturning bound for localized obstacles and a first-order cancellation result for periodic small-amplitude roughness; and
- (vi) formulate a direct trajectory diagnostic for full-Stokes models of Allan Hills and other old-ice sites.

The analysis builds directly on established results. The monotonic-shear no-overturning 95 condition is due to Parrenin and Hindmarsh (2007); the order factor derived below is the reciprocal of the vertical thinning function developed by Parrenin (2013) and implemented by Parrenin et al. (2025); the local wrinkle criterion and finite-strain interpretation are due to WBA and Jacobson and Waddington (2004); and rough-bed extrusion is established (Gudmundsson, 1997a). Here, we write the sign of the inverse thinning

factor as an exact positive–negative u_z/u^3 shear budget. This form gives a full-Stokes residual threshold, localized-obstacle bounds, a restricted periodic-cancellation result and a trajectory diagnostic. These results extend steady-isochrone theory without introducing a new state variable or a universal no-folding theorem.

2 Previous descriptions of stratigraphic folding

2.1 Passive and active folding

Glacier stratification is a material structure produced at the surface and subsequently deformed by flow. Structural glaciology distinguishes passive deformation, in which layers are markers that do not alter the velocity field, from active folding, in which rheological or anisotropic contrasts feed back on stress and motion (Hudleston, 2015; Jennings and Hambrey, 2021). The distinction matters because a prescribed smooth velocity field can amplify or overturn an existing perturbation without selecting its wavelength, whereas active deformation can generate disturbances and select spatial scales.

Microstructural observations show where stratigraphy and rheology may be mechanically coupled. At GISP2, crystal-orientation stripes coincide with localized shear and grain-scale folding, and spatial variation in stripe density may produce larger viscosity contrasts (Alley et al., 1997). Numerical models predict that anisotropy and convergent flow can produce folds spanning scales from cloudy bands to radar reflectors (Bons et al., 2016, 2025; Zhang et al., 2024). The analysis below instead treats age as a passive scalar in a prescribed velocity field. It gives a necessary kinematic test for folding, but it does not describe the growth of an active instability.

2.2 The WBA local wrinkle criterion

Let $m = \delta z/\delta x$ be the slope of an infinitesimal material-line segment $\delta \mathbf{x}$ in a two-dimensional velocity field (u, w) . The continuum-kinematic relation $D\delta \mathbf{x}/Dt = (\nabla \mathbf{u})\delta \mathbf{x}$ describes the deformation of that segment (Gurtin et al., 2010). Taking the ratio of its vertical and horizontal components gives

$$\frac{Dm}{Dt} = w_x + (w_z - u_x)m - u_z m^2. \quad (1)$$

Equation (1), the Riccati form of the material-line equation used by WBA, is therefore standard continuum kinematics. WBA’s distinctive step was to compare the rotation of a disturbed tangent with the rotation of the local steady isochrone tangent m_0 . A second equal-rotation direction separates disturbances that initially flatten toward m_0 from those that rotate away from it. Their shear number expresses the competition between shear-driven rotation and longitudinal extension.

Equation (1) is local. Applied pointwise along a complete curve, it contains the exact tangent kinematics of that curve. The published WBA stability diagnostic, however, does not by itself determine whether an entire equal-age surface loses its horizontal ordering. WBA emphasized this limitation over rough topography: a disturbed tangent can open relative to the local steady isochrone on the lee side of a bump while merely recovering deformation accumulated on the stoss side (Waddington et al., 2001).

2.3 Finite strain and steady isochrone geometry

Jacobson and Waddington (2004) integrated finite deformation along particle paths rather than relying on a homogeneous instantaneous velocity gradient. This is the appropriate framework for deciding whether an inherited open fold becomes recumbent after a long and spatially variable strain history. Their later divide-migration calculation supplied a concrete source of the inherited perturbation (Jacobson and Waddington, 2005).

A separate line of analysis concerns the geometry of isochrones generated continuously in a steady flow. Parrenin et al. (2006) introduced logarithmic total-flux and normalized-streamfunction coordinates in which particle trajectories are straight characteristics when the vertical velocity shape is horizontally uniform. Hindmarsh et al. (2006) compared streamlines and isochrones over basal topography and showed that they are close when

$$\frac{\dot{a}}{U} \frac{L}{H} \ll 1, \quad (2)$$

where $\dot{a}$ is accumulation, U a horizontal speed, L the forcing wavelength and H the ice thickness. Parrenin and Hindmarsh (2007) generalized the characteristic analysis to horizontally non-uniform velocity profiles and wrote isochrone slope as an instantaneous normalized-streamfunction contribution plus a path-history term. In their notation, the sign of $1 - \alpha$ determines whether neighboring particles preserve their order, and $\alpha > 1$ corresponds to an isochrone that has passed through vertical. Section 3.5 relates $1 - \alpha$ directly to the order Jacobian derived below. Parrenin and Hindmarsh (2007) proved that $\alpha > 1$ cannot occur if horizontal velocity increases upward. Hindmarsh et al. (2009) used these solutions to verify a large-scale numerical age model, and Parrenin et al. (2025) recently implemented the same flux-coordinate framework for age, particle origin and vertical thinning.

Table 1 locates the present analysis within this literature.

Table 1: Relationship between established folding analyses and the present formulation.

Study	Primary question	Diagnostic and scope
Waddington et al. (2001)	Does a pre-existing local wrinkle flatten or rotate toward overturning?	Instantaneous tangent slope relative to a steady reference isochrone.
Jacobson and Waddington (2004)	Does a finite-strain history overturn an injected disturbance?	Deformation gradient integrated along trajectories.
Parrenin et al. (2006); Parrenin and Hindmarsh (2007)	How does steady flow shape isochrones and preserve particle order?	Logarithmic flux coordinates, characteristic age solutions and a path factor; monotonic upward speed precludes vertical isochrones.
Hindmarsh et al. (2006)	Do layers drape or override bed and basal-boundary perturbations?	Perturbation and full-mechanical models; wavelength controls architecture.
Hindmarsh et al. (2009); Parrenin et al. (2025)	How can age and thinning be computed accurately in steady flow tubes?	Eulerian and characteristic solvers for age, origin and vertical thinning.
Gudmundsson (1997a)	Can rough-bed Stokes flow contain local extrusion?	Analytical velocity extrema and reverse vertical shear near sinusoidal relief.
This study	Has cumulative reverse shear erased depositional ordering?	Signed inverse-thinning factor, positive–negative shear budget, full-Stokes threshold, obstacle bounds and trajectory diagnostic.

3 Steady age transport and the order Jacobian

3.1 Setting and assumptions

165 Consider a steady two-dimensional glacier velocity

$$\mathbf{u}(x, z) = \begin{bmatrix} u(x, z) \\ w(x, z) \end{bmatrix} \quad (3)$$

where u and w are the horizontal and vertical velocity components, respectively, and x and z are horizontal and vertical coordinates. The domain is bounded above by the steady surface $z = s(x)$. The kinematic result requires only the following assumptions on the
170 region and trajectories of interest:

(A1) The velocity is continuously differentiable and incompressible,

$$u_x + w_z = 0. \quad (4)$$

(A2) Horizontal motion is one-way,

$$u(x, z) > 0, \quad (5)$$

175 after orienting x downstream.

(A3) Every trajectory considered originates in one upstream surface accumulation zone. The accumulation rate, expressed in vertical ice-equivalent units, satisfies $\dot{a}(x) > 0$ where the trajectory enters the ice.

(A4) Stratigraphic age is a passive, non-diffusive scalar. Its steady value $\tau(x, z)$ satisfies

$$180 \quad \frac{D\tau}{Dt} = \underbrace{\tau_t}_{=0} + u\tau_x + w\tau_z = u\tau_x + w\tau_z = 1, \quad \tau = 0 \quad \text{at deposition.} \quad (6)$$

Equation (6) distinguishes a steady age field from the age of a moving parcel. The local derivative τ_t vanishes at fixed position, while the material age increases at one unit of time per unit time along a trajectory.

An impermeable bed is not required for the local result. Basal melting may terminate
185 some trajectories without altering the derivation before termination. Basal freezing, in contrast, introduces a second material source and requires separate origin labels. Assumption (A2) excludes horizontal stagnation and closed circulation. Assumption (A3) requires special care in a blue-ice area because ice leaves through the local surface ablation zone. Its material origin must instead be traced to an upstream surface accumulation zone
190 or to basal freeze-on. Appendix B gives the corresponding mass-flux construction when density varies.

In a two-dimensional incompressible Stokes solution, mass conservation permits the velocity to be represented by a scalar streamfunction ψ . Curves of constant ψ are streamlines because $\mathbf{u} \cdot \nabla \psi = 0$, and the difference in ψ between two streamlines is their
195 volume flux per unit width. The streamfunction is defined up to an arbitrary additive constant and follows from incompressibility alone; no additional rheological assumption is needed. We choose the orientation

$$u = \psi_z, \quad w = -\psi_x. \quad (7)$$

Combining Equations (6) and (7) gives the exact Jacobian identity

$$200 \quad \begin{vmatrix} \frac{\partial \tau}{\partial x} & \frac{\partial \tau}{\partial z} \\ \frac{\partial \psi}{\partial x} & \frac{\partial \psi}{\partial z} \end{vmatrix} = \tau_x \psi_z - \tau_z \psi_x = u\tau_x + w\tau_z = 1. \quad (8)$$

Thus age and streamfunction form local area-preserving coordinates. Isochrones and streamlines intersect transversely wherever the assumptions hold. This is a classical flow-

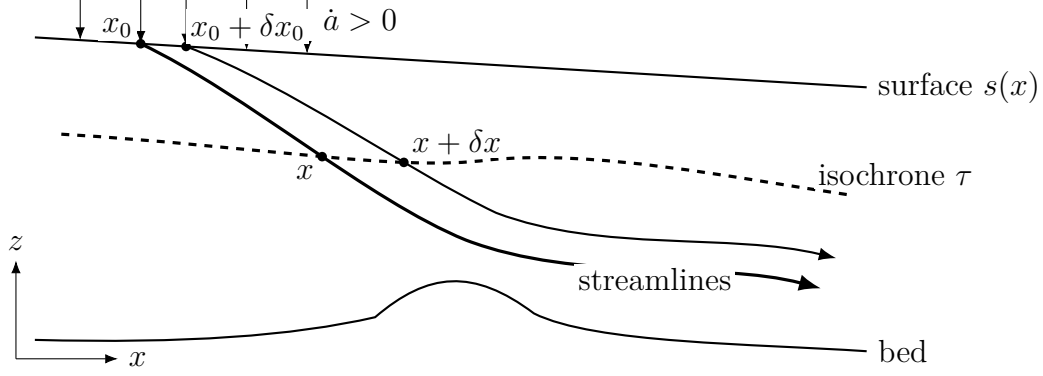


Figure 1: Material map for steady plane flow. Neighboring particles enter through the surface accumulation zone at x_0 and $x_0 + \delta x_0$, follow distinct streamlines and reach positions separated by δx at the same age τ . The order Jacobian is the limit of $\delta x / \delta x_0$. The curves enforce the kinematic requirements used in the derivation—surface origin, non-intersecting streamlines and transverse intersections with the isochrone—but are not a calculated Stokes solution.

coordinate construction; it is useful here because its inverse directly measures stratigraphic order. Appendix A gives the inverse-coordinate identities used below. Figure 1 shows the deposition coordinate, two neighboring trajectories and their positions on an equal-age surface.

3.2 Surface origin and travel time

The streamfunction restricted to the surface is

$$Q(x) = \psi(x, s(x)). \quad (9)$$

The steady surface kinematic condition is

$$u_s s_x - w \Big|_{z=s(x)} = \dot{a}, \quad (10)$$

where $u_s = u(x, s(x))$. Differentiating Q with the chain rule, substituting Equation (7), and then applying Equation (10) gives

$$Q'(x) = \psi_x \Big|_{z=s(x)} + \psi_z \Big|_{z=s(x)} s_x = -w \Big|_{z=s(x)} + u_s s_x = \dot{a}(x) > 0. \quad (11)$$

The increment $d\psi$ is the ice flux per unit width between neighboring streamlines. At the surface, this flux is $\dot{a}_0 dx_0$, so larger accumulation places the same flux interval within a smaller deposition interval. Every streamline label ψ therefore has a unique deposition

position $x_0(\psi)$ satisfying

$$Q(x_0) = \psi, \quad \frac{dx_0}{d\psi} = \frac{1}{\dot{a}_0}, \quad (12)$$

220 where $\dot{a}_0 \equiv \dot{a}(x)|_{x=x_0} = \dot{a}(x_0)$ is the accumulation rate at deposition.

At fixed ψ , Equation (6) gives

$$\left. \frac{\partial \tau}{\partial x} \right|_{\psi} = \frac{1}{u}. \quad (13)$$

Consequently the age at (x, ψ) is the travel time from deposition. Here, ξ is the horizontal coordinate of integration and $z(\xi, \psi)$ is the elevation of the streamline labelled by ψ :

$$225 \quad \tau(x, \psi) = \int_{x_0(\psi)}^x \frac{d\xi}{u[\xi, z(\xi, \psi)]}. \quad (14)$$

At fixed x , Equation (7) implies

$$z_{\psi} = \frac{1}{u}. \quad (15)$$

Stratigraphic order compares travel times on neighboring streamlines, which are neighboring values of ψ . We therefore differentiate Equation (14) with respect to ψ at fixed downstream position. Using Equations (12) and (15) gives

$$230 \quad \tau_{\psi} = -\frac{1}{\dot{a}_0 u_{s0}} - \int_{x_0}^x \frac{u_z}{u^3} d\xi, \quad (16)$$

where $u_{s0} = u[x_0, s(x_0)]$ is the horizontal speed at deposition. The first term records the separation imposed at deposition; the integral records the subsequent differential travel time between neighboring streamlines.

235 3.3 Exact order theorem

Parameterize an isochrone of fixed age by the surface deposition coordinate x_0 . The relevant quantity is not its curvature but the horizontal projection of its tangent.

Definition 1 (Isochrone-order Jacobian). The isochrone-order Jacobian is

$$\mathcal{J}(x, x_0) = \left. \frac{\partial x}{\partial x_0} \right|_{\tau}. \quad (17)$$

240 It measures whether particles deposited in increasing order of x_0 remain in increasing downstream order on an equal-age surface.

Theorem 1 (Cumulative order criterion). *Under assumptions (A1)–(A4), the isochrone-order Jacobian is*

$$\boxed{\mathcal{J} = \frac{u(x, z)}{u_{s0}} \left(1 + \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z}{u^3} d\xi \right)}. \quad (18)$$

245 It is also related exactly to the vertical age gradient by

$$\boxed{\mathcal{J} = -\dot{a}_0 \tau_z.} \quad (19)$$

Hence $\mathcal{J} > 0$ denotes ordered stratigraphy, $\mathcal{J} = 0$ a vertical tangent, and $\mathcal{J} < 0$ an overturned limb.

Proof. At fixed age, differentiation of $\tau(x, \psi)$ gives

$$250 \quad \left. \frac{\partial x}{\partial \psi} \right|_{\tau} = - \frac{\tau_{\psi}}{\tau_x|_{\psi}} = -u\tau_{\psi}. \quad (20)$$

Multiplication by $d\psi/dx_0 = \dot{a}_0$ and substitution of Equation (16) gives Equation (18). At fixed x , $\tau_z = \tau_{\psi}\psi_z = u\tau_{\psi}$, which gives Equation (19). $\square$

The correction of annual-layer thickness for cumulative ice deformation goes back at least to Nye (1963) and Dansgaard and Johnsen (1969). In the present notation, the vertical thinning function λ_z is the current vertical thickness of an infinitesimal age layer divided by its ice-equivalent thickness at deposition. For an ordered layer, a positive age increment $d\tau$ corresponds to the present vertical thickness $-dz = -d\tau/\tau_z$ and the initial thickness $\dot{a}_0 d\tau$. Hence

$$\boxed{\lambda_z = -\frac{1}{\dot{a}_0 \tau_z} = \mathcal{J}^{-1}, \quad \mathcal{J} > 0.} \quad (21)$$

260 Thus $\mathcal{J}$ is not a new independent state variable: in the ordered regime it is the inverse of the established vertical thinning function (Parrenin, 2013; Parrenin et al., 2025). Equation (19) provides its signed continuation through a vertical tangent. The standard positive thinning function diverges as $\tau_z \rightarrow 0$, whereas $\mathcal{J}$ crosses zero regularly and records orientation reversal.

265 At deposition, $x = x_0$, $u = u_{s0}$, and $\mathcal{J} = 1$. At the first vertical tangent, $\tau_z = 0$, but the material curve remains regular. Inverting Equation (8) shows that

$$\left. \frac{\partial z}{\partial x_0} \right|_{\tau} = \dot{a}_0 \tau_x. \quad (22)$$

When $\mathcal{J} = 0$, Equation (6) gives $\tau_x = 1/u$, and therefore

$$\left. \frac{\partial z}{\partial x_0} \right|_{\tau, \mathcal{J}=0} = \frac{\dot{a}_0}{u} > 0. \quad (23)$$

270 The divergence of the conventional slope $m = dz/dx$ at overturning is therefore a coordinate singularity, not a singularity of the layer itself.

3.4 Positive and reverse shear budgets

Define

$$u_z^+ = \max(u_z, 0), \quad u_z^- = \max(-u_z, 0), \quad (24)$$

and the dimensionless cumulative contributions

$$\mathcal{P}(x, x_0) = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^+}{u^3} d\xi, \quad (25)$$

$$\mathcal{R}(x, x_0) = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^-}{u^3} d\xi. \quad (26)$$

Equation (18) becomes

$$\boxed{\mathcal{J} = \frac{u}{u_{s0}} \mathcal{B}, \quad \mathcal{B} \equiv 1 + \mathcal{P} - \mathcal{R}.} \quad (27)$$

The dimensionless factor $\mathcal{B}$ is an order budget: deposition contributes the initial value one, normal shear adds $\mathcal{P}$, and reverse shear subtracts $\mathcal{R}$. Because $u/u_{s0} > 0$, the necessary-and-sufficient condition for preservation of horizontal order is

$$\boxed{\mathcal{R} < 1 + \mathcal{P}.} \quad (28)$$

The first vertical tangent occurs at

$$\boxed{\mathcal{R} = 1 + \mathcal{P},} \quad (29)$$

and overturning occurs when $\mathcal{R} > 1 + \mathcal{P}$.

The unit term in Equations (27)–(29) represents the initial ordering imposed at deposition. Positive u_z , the usual surface-fast glacier profile, adds to the budget. Reverse shear must remove both contributions. Thus a local region with $u_z < 0$ is not itself a fold criterion.

In travel-time coordinates, $dx = u dt$, so

$$\mathcal{R} = \dot{a}_0 u_{s0} \int_0^T \frac{u_z^-}{u^2} dt. \quad (30)$$

where T is the travel time from deposition to the point of interest. Reverse shear is amplified by residence time and by slow horizontal motion. This weighting explains why local extrusion in fast throughflow may leave stratigraphic order unchanged, while modest reverse shear near stagnation can dominate the order budget.

Corollary 1 (Monotonic-shear result). *If $u_z \geq 0$ everywhere along a trajectory, then $\mathcal{R} = 0$, $\mathcal{J} > 0$, and the steady isochrone cannot overturn.*

This corollary is the physical-coordinate statement of the no-overturning result in Appendix A of Parrenin and Hindmarsh (2007). It is sufficient, not necessary. Equation (28)

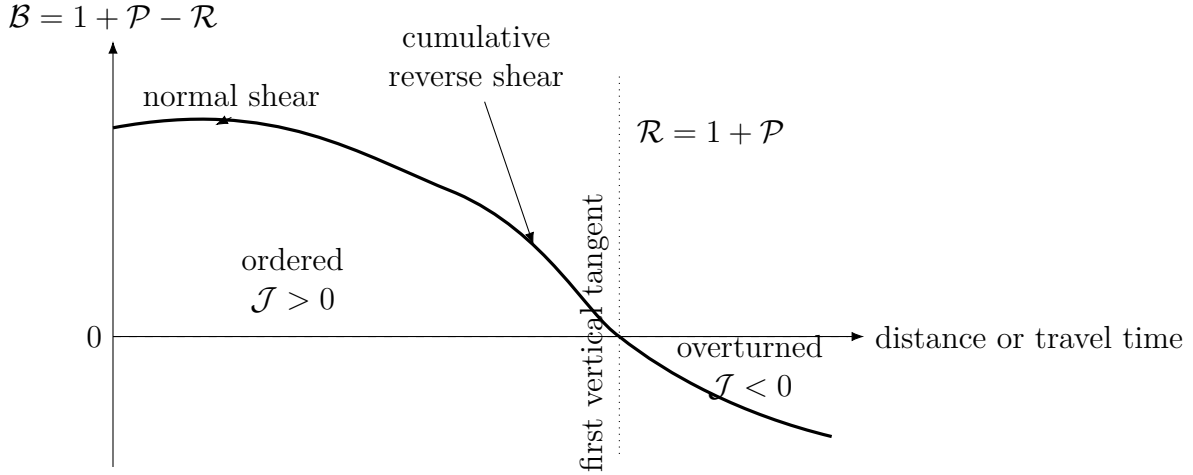


Figure 2: Conceptual order budget. The sign of $\mathcal{B} = 1 + \mathcal{P} - \mathcal{R}$ is the sign of the order Jacobian. Local reverse shear need not cause overturning; a vertical tangent appears only after cumulative reverse shear removes the initial depositional contribution and the accumulated normal-shear contribution.

permits finite intervals of reverse shear provided their cumulative weighted contribution remains below the order budget. Figure 2 summarizes the three contributions and the point at which the budget reaches zero.

3.5 Relation to the Parrenin–Hindmarsh path factor

Parrenin and Hindmarsh (2007) used a normalized streamfunction and logarithmic flux coordinates. For two neighboring particles, their Appendix A gives a mapping of the form

$$\delta\pi = (1 - \alpha) \delta\pi_0, \quad (31)$$

where π_0 and π are their logarithmic flux coordinates at deposition and at the point of interest, respectively, and α is their path-history factor. An isochrone exceeds vertical when $\alpha > 1$. Their present and surface-origin coordinates are monotone functions of x and x_0 wherever assumptions (A2)–(A3) hold. Consequently,

$$\text{sgn}(1 - \alpha) = \text{sgn}\left(\frac{\partial x}{\partial x_0}\bigg|_{\tau}\right) = \text{sgn}(\mathcal{J}) = \text{sgn}(-\tau_z). \quad (32)$$

The present result is therefore not an independent no-overturning theorem. Nor is $\mathcal{J}$ independent of the established thinning variable: Equation (21) identifies it as the signed inverse vertical-thinning factor. The contribution of the physical-space representation is that it exposes the positive and negative parts of vertical shear, remains regular at the first vertical tangent and can be evaluated directly from a numerical velocity field. The `age_flow_line` numerical age model of Parrenin et al. (2025) assumes strictly positive accumulation for its horizontal flux coordinate and no reverse horizontal flow for its vertical

flux coordinate; it computes age and positive vertical thinning within that admissible regime.

4 Full-Stokes constraints on isochrone order

320 4.1 Exact stress decomposition of vertical shear

The order theorem is kinematic. To determine whether glacier mechanics forces Equation (28) to hold, consider steady, isothermal or thermomechanically prescribed, isotropic generalized-Newtonian Stokes flow,

$$\nabla \cdot \mathbf{u} = 0, \quad (33)$$

$$-\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{g} = \mathbf{0}, \quad (34)$$

$$\boldsymbol{\tau} = 2\eta \mathbf{D}, \quad \mathbf{D} = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^\top), \quad \eta > 0. \quad (35)$$

Here, p is pressure, ρ is ice density, $\mathbf{g}$ is gravitational acceleration, $\boldsymbol{\tau}$ is deviatoric stress, $\mathbf{D}$ is the strain-rate tensor and $\eta > 0$ is the scalar effective viscosity. The viscosity may depend on temperature and effective strain rate. Anisotropic rheology is discussed separately below.

325 With z vertical and gravity having no horizontal component, the horizontal momentum equation is

$$-p_x + \tau_{xx,x} + \tau_{xz,z} = 0. \quad (36)$$

Write

$$p = p_{\text{atm}} + \rho g(s - z) + p_{\text{nh}}, \quad (37)$$

330 where $g = |\mathbf{g}|$, p_{atm} is atmospheric pressure and p_{nh} contains all nonhydrostatic pressure, including the finite-slope surface correction. Integrating Equation (36) downward from the free surface, with ζ as the vertical coordinate of integration, gives

$$\tau_{xz}(x, z) = \tau_{xz}^s - \rho g s_x (s - z) + \int_z^s (\tau_{xx,x} - p_{\text{nh},x}) \, d\zeta. \quad (38)$$

Because $\tau_{xz} = \eta(u_z + w_x)$,

$$335 \quad \boxed{u_z = \underbrace{\frac{\rho g(s - z)(-s_x)}{\eta}}_G + \underbrace{\left[-w_x + \frac{\tau_{xz}^s}{\eta} + \frac{1}{\eta} \int_z^s (\tau_{xx,x} - p_{\text{nh},x}) \, d\zeta \right]}_{R_{\text{FS}}}}. \quad (39)$$

where the superscript s denotes surface stress. For a surface descending downstream, $s_x \leq 0$, the gravity-driven term G is nonnegative and favors order preservation. The residual R_{FS} contains the full-Stokes effects omitted from the shallow-ice shear balance:

gradients of vertical velocity, longitudinal stress, nonhydrostatic pressure and the Cartesian
 340 surface shear-stress component.

Substitution into Equation (18) gives

$$\mathcal{J} = \frac{u}{u_{s0}} (1 + \mathcal{G} + \mathcal{F}), \quad (40)$$

with

$$\mathcal{G} = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{G}{u^3} d\xi \geq 0, \quad (41)$$

$$\mathcal{F} = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{R_{\text{FS}}}{u^3} d\xi. \quad (42)$$

The exact full-Stokes condition for a vertical or overturned isochrone is

$$\boxed{\mathcal{F} \leq -(1 + \mathcal{G})}. \quad (43)$$

345 A conservative sufficient condition for no overturning is

$$\boxed{\dot{a}_0 u_{s0} \int_{x_0}^x \frac{[R_{\text{FS}}]_-}{u^3} d\xi < 1}. \quad (44)$$

This inequality discards all positive contributions, including gravity-driven shear, and is therefore stronger than necessary but easy to test.

4.2 There is no universal full-Stokes sign theorem

350 Neither incompressibility nor the Stokes equations fixes the sign of u_z . The constitutive relation constrains $u_z + w_x$, while longitudinal stress and nonhydrostatic pressure can alter the Cartesian vertical gradient of horizontal speed. Analytical rough-bed solutions predict local velocity maxima and minima, and therefore regions with $u_z < 0$, even for gravity-driven linear viscous flow (Gudmundsson, 1997a). The historical extrusion-flow
 355 literature likewise shows that surface-fast velocity profiles are not a pointwise theorem of full-Stokes mechanics (Waddington, 2010).

At stronger roughness, the one-way-flow assumption itself can fail. Gudmundsson (1997b) found separated secondary circulation in troughs of sufficiently steep sinusoidal beds. Ideal no-slip Stokes flow near a sharp corner can contain nested Moffatt eddies
 360 (Moffatt, 1964); Meyer and Creyts (2017) extended this mechanism to Glen-law ice in subglacial mountain valleys and showed that oblique flow can produce three-dimensional eddies. These examples are not direct models of Allan Hills: glacier beds may slide, Glen flow is nonlinear, and real corners are rounded. Their theoretical significance is that steady creeping-flow equations do not forbid recirculation. When $u = 0$ or trajectories close,

365 the downstream travel-time coordinate and unique surface-origin map cease to exist, so Equations (18)–(28) no longer apply globally.

The distinction between local extrusion and cumulative order reversal is central. Published rough-bed solutions establish that u_z can be negative. They do not, by that fact alone, establish that $\mathcal{R}$ reaches $1 + \mathcal{P}$. The latter is an order-one integrated threshold.

370 5 Regimes that preserve order

5.1 Shallow-ice flow

Under the conventional shallow-ice approximation,

$$\tau_{xz} \simeq -\rho g(s - z)s_x. \quad (45)$$

375 For downstream flow with $s_x < 0$, $\tau_{xz} > 0$. An isotropic Glen-type rheology gives u_z with the same sign as τ_{xz} , while leading-order basal sliding contributes a depth-independent horizontal speed (Cuffey and Paterson, 2010). Therefore

$$u_z \geq 0, \quad \mathcal{R} = 0, \quad \mathcal{J} > 0. \quad (46)$$

A fixed bed obstacle can produce arches, troughs and finite curvature in shallow-ice stratigraphy, but it cannot overturn a steady isochrone under these assumptions.

380 5.2 Regular thin-film full-Stokes flow

Let

$$\varepsilon = \frac{H}{L} \ll 1 \quad (47)$$

385 be the aspect ratio of a smooth glacier. Higher-order glacier models reproduce the Stokes solution to second order in ε under their regular asymptotic assumptions, including a range of basal slip regimes (Blatter, 1995; Schoof and Hindmarsh, 2010). Suppose the non-shallow contribution scales as

$$R_{\text{FS}} = O\left(\varepsilon^2 \frac{U}{H}\right) \quad (48)$$

through a region of length $L = H/\varepsilon$, and suppose u remains comparable to U . Then

$$\mathcal{F} = O\left(\frac{\dot{a}}{U}\varepsilon\right). \quad (49)$$

Thus small regular full-Stokes corrections cannot erase the initial depositional contribution when

$$\frac{\dot{a}}{U}\varepsilon \ll 1. \quad (50)$$

This is an asymptotic sufficient regime, not a uniform theorem. The approximation can break down near divides, abrupt sliding transitions, contact zones, steep obstacles, very slow flow or other regions where the thin-film expansion is singular or nonuniform.

5.3 A sufficient bound for a localized obstacle

The cumulative criterion yields a direct obstacle bound without requiring pointwise monotonic shear.

Proposition 1 (Localized reverse-shear bound). *Suppose reverse shear is confined to a trajectory segment of horizontal length ℓ . On that segment, assume*

$$u \geq U_{\min} > 0, \quad u_{s0} \leq U_{\text{dep}}, \quad u_z^- \leq \dot{\gamma}_{\text{rev}}, \quad (51)$$

where $U_{\min}$ is the minimum horizontal speed on the segment, U_{dep} bounds the horizontal speed at deposition, and $\dot{\gamma}_{\text{rev}}$ bounds the reverse vertical shear rate. Then

$$\boxed{\mathcal{R} \leq \frac{\dot{a}_0 U_{\text{dep}} \dot{\gamma}_{\text{rev}} \ell}{U_{\min}^3}}. \quad (52)$$

A sufficient condition for no overturning is that the right-hand side be less than one.

Proof. Substitute Equation (51) into Equation (26) and bound the integral by its maximum integrand multiplied by ℓ . $\square$

For an isolated obstacle whose horizontal and vertical influence scales are both $O(H)$, with non-stagnant throughflow,

$$\mathcal{R} = O\left(\frac{\dot{a}}{U}\right). \quad (53)$$

In a typical interior or flank setting where $U \gg \dot{a}$, an isolated fixed bump can create local extrusion and visible draping while remaining far from the reverse-shear value required for a vertical tangent. More generally, setting the right-hand side of Equation (52) equal to one and solving for length gives

$$\ell \gtrsim \frac{U_{\min}^3}{\dot{a}_0 U_{\text{dep}} \dot{\gamma}_{\text{rev}}}. \quad (54)$$

Slow throughflow reduces this required length through the $U_{\min}^3$ numerator. For scale, take $\dot{a}_0 = 0.05 \text{ m yr}^{-1}$, $U_{\text{dep}} = 10 \text{ m yr}^{-1}$, $\dot{\gamma}_{\text{rev}} = 0.01 \text{ yr}^{-1}$ and $\ell = 1 \text{ km}$. Equation (52) gives $\mathcal{R} \leq 0.005$ when $U_{\min} = 10 \text{ m yr}^{-1}$, which rules out overturning under these assumed bounds. With $U_{\min} = 1 \text{ m yr}^{-1}$, the bound is $\mathcal{R} \leq 5$ and no longer rules it out. These

420 illustrative values are not estimates for Allan Hills; they show that a tenfold reduction in minimum speed raises the bound by a factor of one thousand.

Equation (52) makes precise why “flow over a bump cannot fold an isochrone” is often a good glacier approximation but not a universal Stokes theorem. The approximation depends on non-stagnant flow, a localized deformation zone and a small accumulation-to-
425 speed ratio.

5.4 First-order cancellation over periodic roughness about plug flow

A second reason that fixed roughness may have weak cumulative effect is cancellation, but the result requires a plug-flow background. Let

$$430 \quad u(x, z; \epsilon_b) = U_b + \epsilon_b u_1(x, z) + O(\epsilon_b^2), \quad U_b > 0, \quad (55)$$

where ϵ_b is a small amplitude parameter, $U_b > 0$ is spatially uniform, u_1 is periodic in x , and k is its wavenumber. The perturbation has wavelength $2\pi/k$ and zero horizontal mean at each elevation. Along the unperturbed trajectory $z = z_0$,

$$\frac{u_z}{u^3} = \epsilon_b \frac{u_{1,z}(x, z_0)}{U_b^3} + O(\epsilon_b^2). \quad (56)$$

435 Therefore, over one complete wavelength,

$$\int_x^{x+2\pi/k} \frac{u_z}{u^3} d\xi = O(\epsilon_b^2). \quad (57)$$

Appendix C gives the expansion leading to Equation (57). The perturbative frictionless rough-bed solution of Gudmundsson (1997a) has this plug-flow structure: local extrusion occurs, but its first-order stoss- and lee-side contributions cancel over a complete symmetric
440 wavelength.

Cancellation is not guaranteed for a sheared background, a solitary asymmetric obstacle, a finite path ending mid-cycle, strongly nonlinear roughness or near-stagnant trajectories. Equation (57) is therefore a restricted asymptotic result, not a general symmetry theorem.

6 Overturning mechanisms and limits of the assumptions

445

The theorem does not fail when assumptions (A1)–(A4) hold. Instead, it predicts whether the order budget remains positive or reaches the vertical-tangent threshold. This section first describes overturning within the theorem and then identifies settings in which one

or more assumptions are not satisfied, so that the two-dimensional steady coordinate
 450 construction does not apply.

6.1 Overturning within the theorem: cumulative extrusion

The direct steady-flow mechanism is cumulative reverse shear. A glacier with $u > 0$ can overturn a steady isochrone if reverse shear is sufficiently strong, sufficiently persistent or sufficiently concentrated where u is small. Equation (29) gives the exact threshold:

$$455 \quad \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^-}{u^3} d\xi = 1 + \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^+}{u^3} d\xi. \quad (58)$$

A change in boundary conditions is not mechanically necessary if a steady full-Stokes velocity field satisfies this equality. This outcome is predicted by the theorem rather than representing a failure of it. The extrusion-flow literature establishes the necessary local reverse shear but does not evaluate this cumulative isochrone-order threshold.

460 6.2 Stagnation, separation and closed circulation

If u reaches zero, the x -travel-time representation becomes singular. If a secondary circulation forms, a streamline may not have a unique surface origin and may intersect an equal-age surface in a topology not represented by a single downstream coordinate. Strong roughness, cavities and sharp corners are therefore not merely large values of $\mathcal{R}$; they
 465 can invalidate the coordinate system on which the theorem rests. Numerical calculations should treat $u = 0$ as an event and switch to a more general Lagrangian deformation analysis.

6.3 Surface ablation and sublimation

Newly deposited material enters the glacier through a surface accumulation zone with
 470 $\tau = 0$. Existing isochrones terminate where ice emerges in a surface ablation zone. In a blue-ice area, the surface streamfunction need not be globally monotone because

$$Q'(x) = \dot{a}(x) \quad (59)$$

changes sign between accumulation and ablation zones. A streamline may enter through an upstream accumulation zone and emerge through a downstream ablation zone. The
 475 theorem can still be applied along an interior trajectory if that trajectory has a unique origin in the upstream accumulation zone, but Q^{-1} must be constructed only over that part of the surface. Multiple surface accumulation zones or basal freeze-on require a more general material-source label.

Steady ablation most naturally produces upward-bent and truncated layers, not an intact fold. However, migration, onset or cessation of an ablation zone changes the age field relative to the new steady solution and supplies the inherited disturbance considered by WBA. Surface sublimation followed by burial can also preserve an unconformity with steep or truncated limbs. At Allan Hills, present ablation terminates trajectories, whereas changes in the position of the ablation zone introduce time dependence that lies outside the steady theorem.

6.4 Basal melting and freeze-on

Basal melting removes trajectories and truncates layers. Basal freeze-on adds ice at the bed, so one surface deposition coordinate cannot label the entire stratigraphy. Numerical models show that freeze-on can generate plume-like structures and complex internal layering extending through a large fraction of the ice thickness (Leysinger Vieli et al., 2018). A moving melt/freeze boundary additionally makes the velocity and age problem time dependent.

6.5 Time-dependent geometry and basal traction

If the bed, free surface, divide position or basal traction changes, the inherited age field is not the steady age solution of the new velocity field. The perturbation then obeys a homogeneous advection equation only after the change; it need not vanish initially. Divide migration converts a formerly steady Raymond arch into a fold seed (Jacobson and Waddington, 2005). Traveling slippery or sticky patches produce moving-frame critical layers and thickness-scale overturning (Wolovick et al., 2014; Wolovick and Creyts, 2016). Bedrock uplift, rotation or erosion similarly changes the reference stratigraphy and can inject a finite-amplitude disturbance even when subsequent flow is steady.

6.6 Active rheology, anisotropy and three-dimensional flow

Equation (39) assumes isotropic scalar viscosity. Crystal fabric, impurities, grain size and damage can alter both the stress-strain relation and the velocity field in response to stratigraphy. Full-Stokes anisotropic divide models already demonstrate substantial changes in isochrone shape and age-depth relations (Martín et al., 2009a). More strongly coupled heterogeneity can generate folds internally rather than merely advecting an imposed perturbation (Alley et al., 1997; Thorsteinsson and Waddington, 2002; Bons et al., 2016, 2025). The Allan Hills microstructure indicates that anisotropy and localized shear cannot be ignored (Stoll et al., 2026).

The order theorem is also intrinsically two-dimensional. In three dimensions, incompressibility provides a vector potential rather than a scalar streamfunction, and neighboring

surface origins can exchange order in one horizontal projection while remaining separated in the other. Divide models show that along-ridge flow and divide migration alter isochrone architecture even when cross-flow is modest (Martín et al., 2009b). A three-dimensional generalization should track the two-dimensional Jacobian mapping a surface deposition patch to an equal-age surface. The scalar $\mathcal{J}$ is recovered only when cross-flow is negligible and a coherent flow tube exists.

7 Implications for old-ice sites

Allan Hills lies close to several limits in which a two-dimensional fixed-obstacle approximation becomes unreliable. Present-day ice rises into a surface ablation zone above steep nunatak topography (Spaulding et al., 2012; Kehrl et al., 2018; Stoll et al., 2026). The three-dimensional full-Stokes model of Manos et al. (2026) predicts nearly stagnant basal ice upstream of the nunatak, where the bottom 20 m averages approximately $0.4\text{--}0.5\text{ mm yr}^{-1}$. Modeled speed remains nonzero but locally falls to order micrometres per year. In contrast, modeled basal speeds in the Cul-de-sac are approximately $46\text{--}49\text{ mm yr}^{-1}$. These modeled scales distinguish local preservation in slow ice from transport of old ice into the Cul-de-sac.

Near-stagnant flow has two consequences for the present calculation. First, the u^{-3} weighting in Equation (26) becomes large and numerically sensitive even though assumption (A2) may remain technically satisfied. Second, the modeled flow is fundamentally three-dimensional. It contains important local bridging stresses associated with horizontal gradients of vertical velocity; the corresponding $-w_x$ term appears explicitly in the full-Stokes shear decomposition, Equation (39). A two-dimensional order budget is therefore meaningful only within a coherent flow tube for which cross-flow is small. The Cul-de-sac presents a separate provenance problem: its geochemistry and modeled transport indicate a distinct accumulation environment (Manos et al., 2026). The assumption that each trajectory can be represented by one surface accumulation zone is therefore unlikely to describe all Cul-de-sac ice, and a three-dimensional, multi-source material map is needed there.

The structural observations constrain, but do not determine, the deformation history. The 69° mean layer dip in ALHIC2302 is steep but remains below vertical (Kirkpatrick et al., 2025). Centimetre-scale z-folds in ALHIC1901 record local folding, but they do not by themselves explain every core-scale age reversal (Stoll et al., 2026). The inverted age–depth relation in ALHIC1901 more directly indicates that τ_z changes sign over some sampled intervals (Shackleton et al., 2025). Equation (19) then implies $\mathcal{J} < 0$ if the samples lie on one continuous family of surface-deposited layers. Multiple source regions, unconformities or faults could instead produce an apparent age reversal.

The same calculation is relevant to deep old-ice drilling. At the Beyond EPICA site, radar and flow modelling indicate strong basal thinning and possible stagnant or disturbed

ice near the bed (Lilien et al., 2021; Chung et al., 2025). Evaluating $\mathcal{J}$ along model trajectories would test whether the steady velocity field preserves order above this basal unit. The boundary conditions differ from those at Allan Hills, but both settings require a mechanical link between modelled flow and the stratigraphic age interpretation.

For old-ice stratigraphic interpretation, the equations define four tests of whether steady local topography can produce an observed fold or age reversal.

(1) **Steady fixed geometry.** Compute the present or reconstructed steady full-Stokes field and evaluate $\mathcal{P}$, $\mathcal{R}$ and $\mathcal{J}$ along source-to-core trajectories. If the order budget $1 + \mathcal{P} - \mathcal{R}$ remains well above zero, fixed present-day geometry cannot produce the observed recumbent folding.

(2) **Steady but singular or multi-source flow.** Test for stagnation, recirculation, basal freeze-on and surface origin multiplicity. If these occur, the simple order theorem is inapplicable and the full material map must be used.

(3) **Inherited unsteady stratigraphy.** If the steady model preserves order, evaluate changes in ablation pattern, ice thickness, flow direction, divide position, bed geometry and basal traction. These mechanisms inject finite disturbances that can later be amplified by shear.

(4) **Active deformation.** If kinematic models underpredict observed folds, couple fabric and rheological heterogeneity to flow. The microstructural observations make this mechanism plausible at centimetre scale.

This hierarchy distinguishes regimes in which changing boundary conditions are mechanically required. In regular, shallow, non-stagnant flow, Equation (52) keeps the order budget well above zero, so a time-dependent or rheological mechanism is needed. In more general full-Stokes flow, sufficiently cumulative extrusion or loss of one-way flow can overturn steady isochrones without a boundary change.

8 Numerical diagnostic for full-Stokes models

The order criterion can be evaluated without explicitly solving the age equation or advecting a dense family of isochrones. Start trajectories in a surface accumulation zone and record x_0 , $\dot{a}_0$ and u_{s0} . With t denoting time since deposition, integrate along each trajectory

$$\frac{dx}{dt} = u, \quad \frac{dz}{dt} = w, \quad (60)$$

$$\frac{d\mathcal{P}}{dt} = \dot{a}_0 u_{s0} \frac{u_z^+}{u^2}, \quad \frac{d\mathcal{R}}{dt} = \dot{a}_0 u_{s0} \frac{u_z^-}{u^2}, \quad (61)$$

with $\mathcal{P}(0) = \mathcal{R}(0) = 0$. Then

$$\boxed{\mathcal{J}(t) = \frac{u(t)}{u_{s0}} [1 + \mathcal{P}(t) - \mathcal{R}(t)]}. \quad (62)$$

The diagnostic is

$$\min_t [1 + \mathcal{P}(t) - \mathcal{R}(t)] \begin{cases} > 0, & \text{order preserved,} \\ = 0, & \text{first vertical tangent,} \\ < 0, & \text{overtaken steady isochrone.} \end{cases} \quad (63)$$

580 A robust implementation should include four checks. First, compare Equation (62) with $-\dot{a}_0\tau_z$ from an independently solved steady age equation. Second, advect at least one pair of neighboring particles to verify the sign of $\partial x/\partial x_0|_\tau$. Third, stop the downstream-coordinate integration when u approaches zero and diagnose recirculation with the full trajectory topology. Fourth, compute u_z using the finite-element gradient rather than
585 differencing noisy interpolated velocities. Separate storage of $\mathcal{P}$ and $\mathcal{R}$ is preferable to integrating their difference because large compensating contributions can otherwise obscure numerical error.

For transient flow, Equation (62) is not valid because the steady travel-time representation fails. The corresponding calculation should integrate the deformation of the
590 two-parameter deposition map through time. Nevertheless, the steady order budget remains useful as a baseline and as a diagnostic applied to quasi-steady snapshots.

9 Numerical Examples

We use two-dimensional full-Stokes calculations to illustrate both preservation and reversal of isochrone order. These calculations test the mechanics in controlled geometries; they
595 are not reconstructions of Allan Hills or another old-ice site. The first two calculations use a periodic domain that is 800 m long and 160 m thick. Ice obeys Glen’s flow law with $n = 3$, the bed is fixed and no slip, and the flat upper surface is traction free. Gravity acts at a surface slope of 0.03 rad. The rate factor gives a flat-slab surface speed of 0.0138 myr^{-1} , and the effective strain rate is regularized at 10^{-10} yr^{-1} . We
600 solve for velocity and pressure on 48 by 10 structured mesh divisions using Taylor–Hood elements, with quadratic velocity and linear pressure, and obtain the nonlinear solution by continuation through $n = 1, 2, 3$. The surface is held steady by diagnosing the accumulation rate from the kinematic boundary condition, $\dot{a} = -w|_{z=H}$. Trajectories begin only where $\dot{a} > 0$, so that x_0 remains a physical surface-origin coordinate.

605 The first calculation compares a 16 m-high symmetric obstacle with an 80 m-high

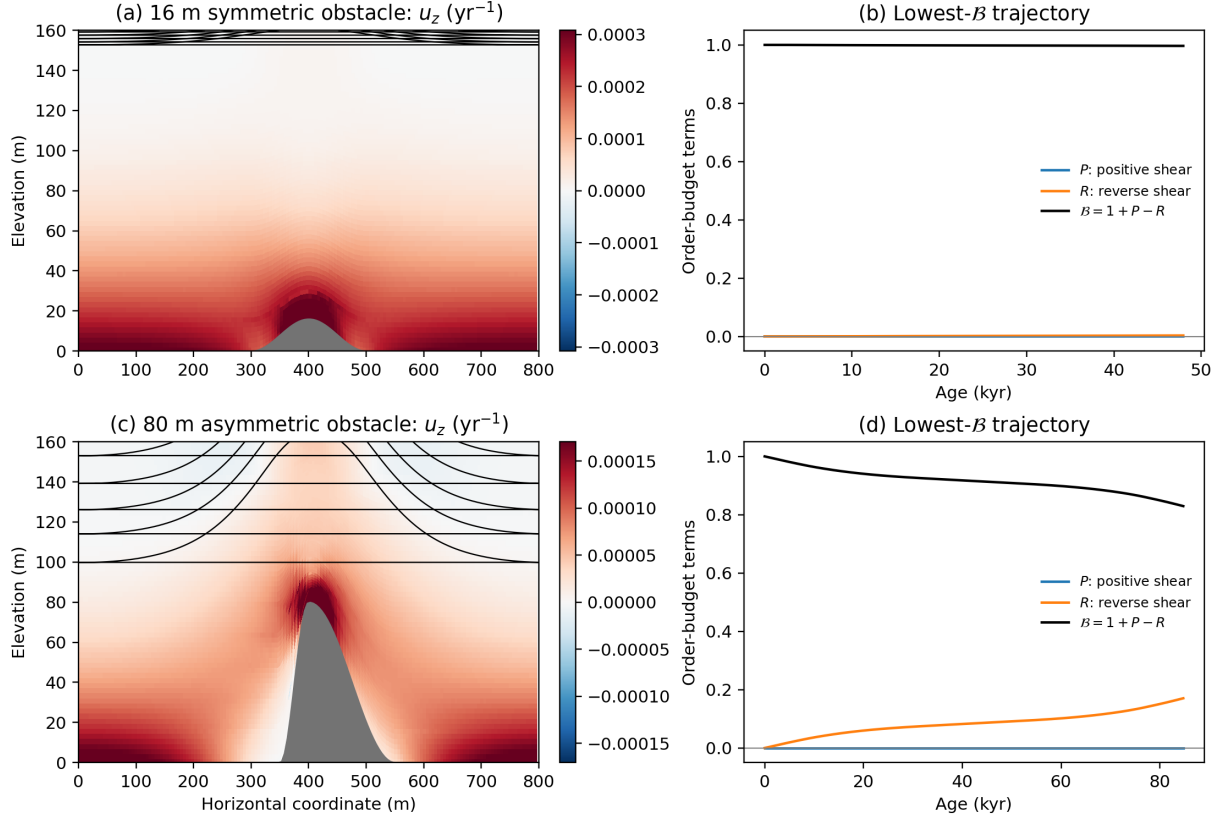


Figure 3: Full-Stokes order-budget examples. The left panels show vertical shear and five surface-origin trajectories for a low symmetric obstacle and a larger asymmetric obstacle. The right panels show the positive-shear contribution $\mathcal{P}$, reverse-shear expenditure $\mathcal{R}$ and remaining order budget $\mathcal{B}$ for the trajectory with the lowest final budget in each case. Local reverse shear is appreciable for the asymmetric obstacle, but $\mathcal{B}$ remains positive.

asymmetric obstacle (Figure 3). Both obstacles have raised-cosine profiles. The symmetric obstacle has 100 m half-widths; the asymmetric obstacle has 50 m and 150 m upstream and downstream half-widths. Five trajectories sample the cumulative positive accumulation flux in each case. The symmetric obstacle produces little reverse-shear expenditure: the minimum value of $\mathcal{B}$ is 0.9966. The asymmetric obstacle produces a broader region of reverse vertical shear. Along the most strongly affected trajectory, $\mathcal{R} = 0.1705$ and $\mathcal{B} = 0.8295$. Thus the obstacle changes the spacing and dip of the layers, but it does not exhaust the order budget or produce a vertical tangent in this calculation.

We also check the exact identity in Equation (27) against a separate numerical calculation. For each trajectory, we estimate $\mathcal{J}$ by tracing equal-age particles from $x_0 \pm 0.05$ m. This finite-difference estimate agrees with $(u/u_{s0})\mathcal{B}$ to a maximum relative error of 6.6×10^{-4} (Figure 4). The comparison tests the trajectory integration and the cumulative shear budget without solving the age equation.

The second calculation narrows the minimum ice thickness above the asymmetric obstacle from 144 to 2 m (Figure 5). We follow the trajectory at 70% of the cumulative positive accumulation flux. As the gap narrows, its minimum horizontal speed decreases

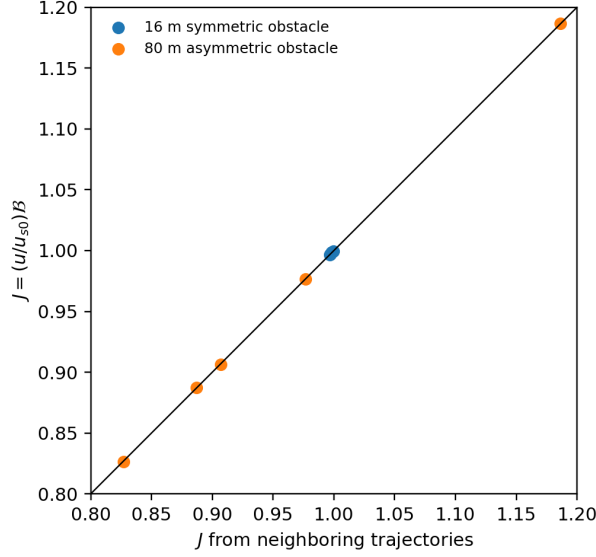


Figure 4: Independent evaluation of the order Jacobian. The horizontal axis uses equal-age trajectories initialized at $x_0 \pm 0.05$ m; the vertical axis uses Equation (27). The line shows equality.

from 0.0128 to 0.0020 m yr^{-1} , its travel time increases from 35 to 178 kyr, and its order budget decreases from 0.998 to 0.705. The reverse-shear contribution grows, but the sampled surface-origin trajectory remains ordered.

The 2 m gap also produces a near-stagnant zone next to the no-slip bed. On meshes with 48, 72 and 96 horizontal elements, the minimum sampled horizontal velocities are -0.423 , -0.037 and -0.022 $\mu\text{m yr}^{-1}$, respectively. The negative magnitude decreases under refinement, so the calculation does not resolve reverse flow or a recirculating eddy. It instead shows that velocity near the obstacle approaches a scale at which the u^{-2} trajectory diagnostic becomes resolution sensitive. We therefore stop the physical interpretation at near-stagnation. Distinguishing slow throughflow from recirculation in this setting would require local mesh refinement and, for an old-ice site, three-dimensional geometry.

The obstacle calculations preserve order and therefore serve as negative controls. A positive control is needed to show the geometry and sign change of an overturned isochrone. We use the two-stage mechanism of Waddington et al. (2001): a transient process first creates a finite wrinkle, after which ordinary surface-fast shear rotates its downstream-dipping limb through vertical. The full-Stokes domain is a 6 km-long and 500 m-thick parallel-sided slab with the same surface slope, no-slip bed, traction-free surface and $n = 3$ rheology. Analytical slab velocities are imposed at the two sides. Calculations on 48 by 8 and 96 by 16 mesh divisions give maximum vertical shear rates of 1.071×10^{-3} and 1.077×10^{-3} yr^{-1} , respectively.

The initial isochrones are non-intersecting sinusoids with 25 m amplitude and 500 m wavelength. Their maximum initial dip is 17.4° . The first material segment reaches $\partial x / \partial X = 0$, and hence a vertical tangent, after 3.2 kyr. At 25 kyr, the fine calculation

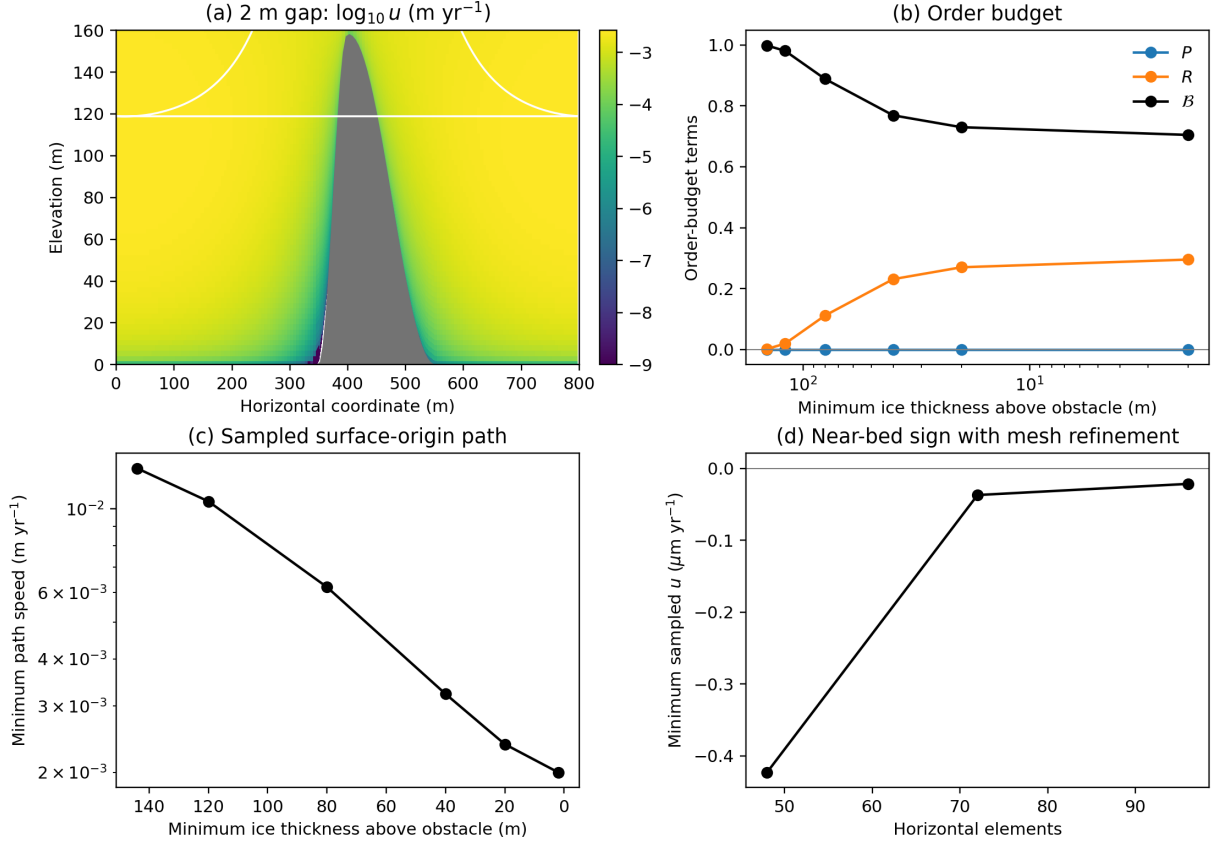


Figure 5: Approach to near-stagnation. (a) Horizontal speed and the sampled surface-origin trajectory for a 2 m gap. (b) Positive shear, reverse shear and the order budget as the gap narrows. (c) Minimum horizontal speed along the surface-origin trajectory. (d) Minimum near-bed velocity sampled on three meshes. The negative values approach zero with refinement and do not establish a resolved flow reversal.

contains 4544 segments with $\partial x/\partial X < 0$, whereas the initially flat control contains none. Figure 6 shows the resulting recumbent isochrons and multiple age reversals along a vertical synthetic core. The numerical age values are schematic; the sign of $\partial x/\partial X$ establishes overturning independently of that age scale.

This positive control does not contradict the steady order theorem. Its sinusoidal isochrones are an inherited disturbance rather than the exact steady age field associated with the slab velocity. The calculation instead illustrates the finite-amplitude fold mechanism that becomes available when time-dependent geometry or another process has already supplied a seed.

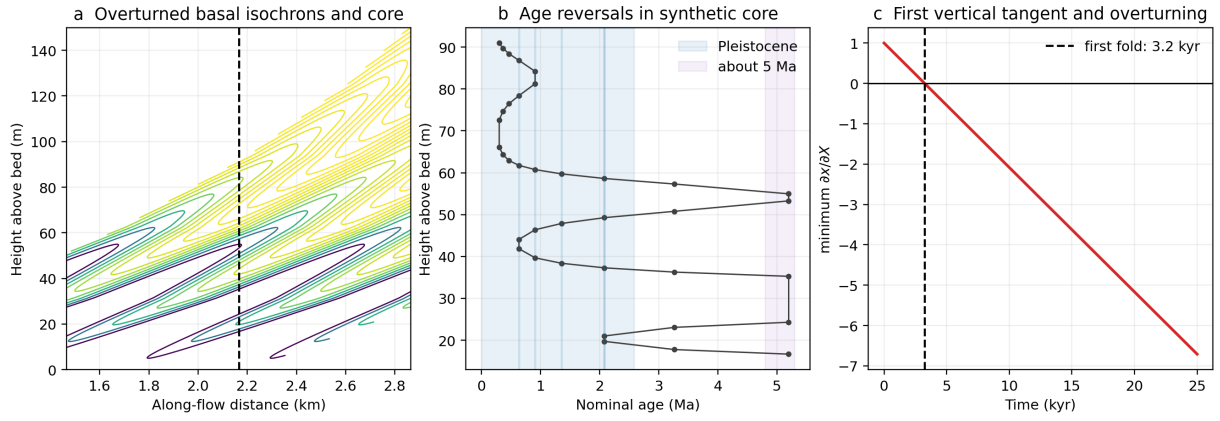


Figure 6: Overturned isochrons in the full-Stokes positive control. (a) The initially ordered isochrons after 25 kyr of advection through the $n = 3$ slab velocity field. The downstream-dipping limbs have rotated through vertical and become recumbent. The dashed line marks the synthetic core. (b) Repeated intersections of the overturned isochrons produce age reversals in the core. The nominal ages are schematic. (c) The minimum horizontal material derivative reaches zero after 3.2 kyr and is negative thereafter. This sign change, rather than layer relief alone, diagnoses overturning.

10 Discussion

10.1 A theorem about deposition order, not about curvature

The curvature equation for an advected material line contains first and second spatial derivatives of velocity. It correctly shows that an obstacle can create curvature from an initially straight layer. The present order criterion answers a different question. A recumbent fold requires the horizontal projection of an equal-age surface to reverse, or equivalently τ_z to change sign. An obstacle may generate large curvature while $\mathcal{J}$ remains positive everywhere. This is the rigorous sense in which a steady bump can disturb isochrones without supplying the order-reversing seed required for a recumbent fold.

WBA's statement that steady-state layers are not passively folded can now be made conditional. If a unique steady age field exists and Equation (28) holds, exact steady stratigraphy is an invariant solution and remains ordered. A prescribed steady flow cannot spontaneously create a perturbation relative to that solution. However, arbitrary material lines placed in the same velocity field are not subject to this restriction, and the conclusion does not hold if cumulative reverse shear crosses the threshold or if the surface-origin coordinate ceases to be global.

10.2 Why local extrusion is usually insufficient

Full-Stokes solutions contain local extrusion, but local extrusion does not imply order reversal. Equation (27) can be read as an order budget. Deposition supplies an initial contribution of one, surface-fast shear adds $\mathcal{P}$, and reverse shear subtracts $\mathcal{R}$. Reverse shear around a localized or periodic obstacle may also be short-lived or cancel between stoss and lee sides. Equation (52) bounds this subtraction, while Equation (57) gives the perturbative cancellation.

Near stagnation, the reverse-shear subtraction becomes much more sensitive to speed. The u^{-3} spatial weighting and u^{-2} temporal weighting imply that a slowly moving basal zone can accumulate a large order change even when the pointwise shear is moderate. This is also the regime in which age becomes large, numerical conditioning deteriorates and recirculation becomes possible. A steady-obstacle explanation for Allan Hills folding must therefore identify a slow reverse-shear trajectory and show that the order budget reaches zero in Equation (29); showing $u_z < 0$ is insufficient.

10.3 Boundary changes as fold seeds

Time-dependent boundary processes provide several mechanisms for creating finite inherited disturbances. Surface sublimation can truncate and steepen layers; migration of an ablation zone can leave that geometry to be buried or sheared. Basal melt and freeze-on

alter material sources and sinks. Divide migration changes the reference velocity profile. Traveling traction anomalies create a moving-frame critical layer and material trapping. Bedrock uplift or rotation changes both geometry and stress. Each mechanism breaks at least one assumption behind the steady order theorem and can therefore create a nonzero perturbation relative to the steady stratigraphy.

The order framework does not identify which mechanism operated at Allan Hills. It supplies a rejection test. If a high-resolution, history-consistent full-Stokes model of fixed geometry keeps the order budget well above zero on all relevant trajectories, then steady topographic draping is excluded as the cause of recumbent age reversals. The remaining explanation must invoke time dependence, multiple material sources, active rheology, three-dimensional exchange or structural discontinuity.

10.4 Novelty and relationship to established theory

Several components of this manuscript are established. Equation (8) is a standard consequence of steady incompressible transport. Logarithmic flux coordinates and characteristic age solutions were developed by Parrenin et al. (2006) and Parrenin and Hindmarsh (2007), and the latter paper contains both the monotonic-shear corollary and the path dependence of isochrone slope. The vertical thinning function was derived in the same framework by Parrenin (2013) and is now computed operationally by Parrenin et al. (2025); Equation (21) shows that the present order factor is its inverse in the ordered regime. The need for a transient wrinkle and the competition between shear and extension are due to Waddington et al. (2001). Finite-strain amplification is treated by Jacobson and Waddington (2004). Local extrusion and rough-bed recirculation are known full-Stokes phenomena (Gudmundsson, 1997a,b).

The analysis therefore reformulates established age–streamfunction and thinning theory for use with numerical velocity fields. Its distinct result is the physical-space identity that writes signed inverse thinning as a cumulative u_z/u^3 integral. The positive–negative decomposition in Equations (25)–(28) then gives the full-Stokes residual threshold in Equation (43), the localized-obstacle bound in Equation (52), the plug-flow cancellation result in Equation (57) and the split trajectory diagnostic in Equation (62). These results are a cautious extension of Parrenin–Hindmarsh theory rather than a universal theorem about incompressible Stokes flow.

The idealized calculations in Section 9 verify the diagnostic and show the recumbent geometry that follows from a sign reversal, but they do not test Allan Hills geometry. The next step is to apply the diagnostic to reconstructed geometry and flow history and to identify a physical source for any inherited disturbance. An order budget that remains well above zero would exclude fixed roughness as the cause of overturning, whereas a trajectory crossing $\mathcal{J} = 0$ would identify where steady deformation is sufficient. Layer attitudes, age

725 reversals and fabric provide independent observations against which to evaluate either
result.

10.5 Limitations

The analysis neglects diffusion of age markers, fracture and discontinuous displacement. It
assumes a smooth two-dimensional velocity field and that each trajectory originates in
730 one surface accumulation zone. The full-Stokes decomposition assumes isotropic scalar
viscosity. It does not establish a universal mechanical bound on R_{FS} , because none follows
from the Stokes equations alone. The localized-obstacle estimate depends on externally
supplied bounds on minimum speed, deposition speed and reverse shear. The periodic
cancellation result is first order and can be destroyed by asymmetry or nonlinear geometry.

735 Most importantly, the theorem diagnoses overturning of a continuous material isochrone.
An observed age reversal can also result from faulting, unconformity, multiple source regions,
freeze-on ice, sampling across a sharply dipping but discontinuous structure, or unresolved
three-dimensional geometry. The order diagnostic should therefore be used with structural
observations rather than as a stand-alone interpretation.

740 11 Conclusions

1. A steady two-dimensional isochrone preserves depositional order if and only if its
signed order Jacobian remains positive (cf. Equation (18)). The equivalent observable
condition is $\tau_z < 0$ (Equation (19)), and in the ordered regime $\mathcal{J}$ is the reciprocal of
the established vertical thinning function (Equation (21)).
- 745 2. Pointwise surface-fast shear, $u_z \geq 0$, is sufficient but not necessary. The exact condition
is the cumulative order budget $\mathcal{R} < 1 + \mathcal{P}$ (Equation (28)). Local extrusion changes
stratigraphic order only if its weighted path integral removes the initial depositional
contribution and all accumulated normal shear.
3. Full-Stokes mechanics does not guarantee this condition for arbitrary glacier geometry.
750 Gravity-driven shear favors order, but longitudinal stress, nonhydrostatic pressure and
horizontal gradients of vertical velocity can generate reverse shear (cf. Equation (39)).
Stagnation and recirculation invalidate the one-way coordinate construction entirely.
4. The shallow-ice regime preserves order (Equation (46)). A non-stagnant localized
obstacle obeys the speed-and-shear bound in Equation (52), and small-amplitude
755 periodic roughness about a uniform plug-flow background has no first-order net order
change over a complete wavelength (Equation (57)).

5. The idealized full-Stokes examples are consistent with these predictions. An asymmetric obstacle reduces the sampled order budget to 0.8295, while narrowing the remaining ice gap to 2 m reduces it to 0.705 (Figures 3–5). Neither steady-obstacle calculation produces a vertical tangent. In contrast, a 17.4° inherited wrinkle reaches a vertical tangent after 3.2 kyr and develops overturned isochrons under ordinary surface-fast shear (Figure 6).
6. The theory converts the intuition that fixed obstacles produce draping rather than folds into a conditional, testable statement. In ordinary fast, shallow throughflow, boundary or rheological changes are effectively required to seed recumbent folding. In slow, rough, near-stagnant full-Stokes flow, fixed geometry remains a viable mechanism until the cumulative diagnostic in Equation (62) is evaluated.
7. For NSF COLDEX at Allan Hills, the next calculation is to integrate $\mathcal{P}$, $\mathcal{R}$ and $\mathcal{J}$ along reconstructed source-to-core trajectories using Equations (60)–(62). A three-dimensional material map is required where cross-flow or multiple source regions are important. The same diagnostic can test stratigraphic order at continuous old-ice sites such as Beyond EPICA.

Data availability

No new observational data are presented. The analytical results are fully specified by the equations in this manuscript. The numerical diagnostic requires a model velocity field and its vertical derivative.

Code availability

Three executed notebooks in the `notebooks` directory accompany this manuscript. They contain the full-Stokes solves, trajectory calculations, numerical checks and figure generation.

Author contribution

Bradley Paul Lipovsky developed the analysis and wrote the manuscript draft.

Competing interests

To be completed by the author before submission.

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A Inverse-coordinate identities

Equation (8) can be written

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$$\begin{bmatrix} \tau_x & \tau_z \\ \psi_x & \psi_z \end{bmatrix} \begin{bmatrix} dx \\ dz \end{bmatrix} = \begin{bmatrix} d\tau \\ d\psi \end{bmatrix}, \quad \det = 1. \quad (64)$$

The inverse transformation is

$$\begin{bmatrix} dx \\ dz \end{bmatrix} = \begin{bmatrix} \psi_z & -\tau_z \\ -\psi_x & \tau_x \end{bmatrix} \begin{bmatrix} d\tau \\ d\psi \end{bmatrix} = \begin{bmatrix} u & -\tau_z \\ w & \tau_x \end{bmatrix} \begin{bmatrix} d\tau \\ d\psi \end{bmatrix}. \quad (65)$$

Therefore

$$x_\tau|_\psi = u, \quad z_\tau|_\psi = w, \quad x_\psi|_\tau = -\tau_z, \quad z_\psi|_\tau = \tau_x. \quad (66)$$

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The first two identities state that advancing one unit in age at fixed streamline follows the velocity for one unit of time. The latter two specify the equal-age tangent. Multiplication by $d\psi/dx_0 = \dot{a}_0$ yields Equation (19) and the corresponding vertical tangent component.

B Extension to steady variable-density flow

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Below the firn, glacier ice is accurately treated as incompressible with nearly constant density. The coordinate construction extends to a steady variable-density flow satisfying

$$\nabla \cdot (\rho \mathbf{u}) = 0. \quad (67)$$

Define the mass-flux streamfunction by

$$\rho u = \psi_z, \quad \rho w = -\psi_x. \quad (68)$$

Then

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$$\begin{vmatrix} \tau_x & \tau_z \\ \psi_x & \psi_z \end{vmatrix} = \rho(u\tau_x + w\tau_z) = \rho. \quad (69)$$

The mapping remains nonsingular for $\rho > 0$, and the same order interpretation follows after replacing volume flux by mass flux. The constant-density formula used in the main text is therefore not conceptually tied to strict incompressibility, although the weights in the cumulative integral acquire density factors.

C First-order periodic cancellation about plug flow

Let

$$u(x, z; \epsilon_b) = U_b + \epsilon_b u_1(x, z) + O(\epsilon_b^2), \quad U_b > 0, \quad (70)$$

where U_b is spatially uniform and u_1 is periodic in x with wavelength $2\pi/k$ and zero horizontal mean at each elevation. Then

$$u_z = \epsilon_b u_{1,z} + O(\epsilon_b^2), \quad u^{-3} = U_b^{-3} + O(\epsilon_b). \quad (71)$$

Because the background integrand is identically zero, the $O(\epsilon_b)$ displacement of the trajectory affects the integral only at $O(\epsilon_b^2)$. Evaluating the first-order term along the unperturbed streamline $z = z_0$ gives

$$\int_x^{x+2\pi/k} \frac{u_z}{u^3} d\xi = \frac{\epsilon_b}{U_b^3} \int_x^{x+2\pi/k} u_{1,z}(\xi, z_0) d\xi + O(\epsilon_b^2) = O(\epsilon_b^2), \quad (72)$$

because differentiation with respect to z preserves the zero horizontal mean of u_1 . This establishes Equation (57). For a vertically sheared background, trajectory displacement can act on the nonzero background integrand at first order, so the cancellation does not follow without additional assumptions.

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